Overview

The code in this replication package installs all necessary commands and runs all the analyses for the paper " From High School to Higher Education: Is recreational marijuana a consumption amenity for US college students?" by Ahmed El Fatmaoui. All analysis is done in R. A parent file (*master.R*) can be used to run all files at once, calling other scripts to install and load required packages and create tables and figures. master.R script code saves tables in tables folder and figures in figures folder. To replicate a single table or figure, run only the source named after that table or figure (e.g., run source("*table_A7.R*") to replicate table A7). Replicators can expect the code to take about 1 hour to run. Mainly, replication of figure 6 and table A9 takes a longer time to run as it computes the distance between each college location and the closest treated state border.

Figure 1: Layout of Replication Files

```
replication-files-rml
├─ data
│   ├─ clean_data (cleaned/merged data)
│   └─ source_data (raw data)
│       ├─ controls
│       ├─ ipeds
│       └─ others
├─ programs
│   ├─ data_cleaning (Extracts data from the sources and merges data)
│   │   └─
│   │       ├─ bls_bea
│   │       ├─ census_pop
│   │       ├─ completion_cleaning
│   │       ├─ enrollment_cleaning
│   │       ├─ google_trend
│   │       ├─ grad_rates_panel
│   │       ├─ ipeds_downloading
│   │       └─ marijuana_price
│   └─ fig_tab (creates tables and figures)
│       └─
│           ├─ master.R (performs all data cleaning and analysis procedures)
│           ├─ figure_#.R
│           └─ table_#.R
├─ figures
│   ├─ appendix
│   └─ main
└─ tables
    ├─ appendix
    └─ main
```

Figure 1 illustrates the layout of the replication files. The *data* folder encompasses two primary subfolders. One folder (*source_data*) houses the raw data obtained from National Center for Education Statistics (2022*a*) or other sources (Bureau of Labor Statistics, 2021, Bureau of Economic Analysis, 2021, U.S. Census Bureau, 2021). The other folder contains the processed data, specifically the merged IPEDS data.

Similarly, the *programs* folder, which comprises all the R scripts, includes a subfolder (*data_cleaning*) responsible for downloading and cleaning the data. Another subfolder (*fig_tab*) within the *programs* folder executes all analyses. The latter saves figures and tables in their respective folders located within the last two major folders illustrated in Figure 1.

Data Availability and Provenance Statements

The paper uses mainly IPEDS data National Center for Education Statistics (2022*a*), and two other data from Bureau of Economic Analysis (2021)[1], Bureau of Labor Statistics (2021), and U.S. Census Bureau (2021). Other data used in appendix are from Google Trends and priceofweed.com. See section 3 and appendix B for detailed description of these data. I certify that the author(s) of the manuscript have legitimate access to and permission to use, redistribute, and publish the data used in this manuscript. All data are publicly available and have been deposited in the ICPSR repository of this paper.

Dataset List

Table 1: Table 1: datasets Used in Paper

Dataset	Data files	Description and processing	Data Location	Citation	Provided
IPEDS: Fall Enrollment	enroll_main.csv, enroll_all.csv, enroll_vocational.csv	**Data source** (Link): - IPEDs Survey: Fall Enrollment - IPEDs Title: Race/ethnicity, gender, attendance status, and level of student: Fall (2009-2019). **Data Processing** - *data_cleaning/ipeds_downloading/IPEDS_scraping.R* downloads the data. - *data_cleaning/enrollment_cleaning/enroll_cleaning.R* merges the data together and saves for vocational (enroll_vocational), academic (enroll_main) and all institutions (enroll_all).	data/clean_data	National Center for Education Statistics (2022*a*)	TRUE

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[1]This data is extracted using the bea.R package in R. Prior to accessing the data, the user must obtain an API key. Further details can be found at https://github.com/us-bea/bea.R.

Table 1 – continued from previous page

Dataset	Data files	Description and processing	Data Location	Citation	Provided
IPEDS—Graduation rates	grad_rates.csv	**Data source** (Link): - The link will navigate you to the dataset detailing graduation rates over 4, 5, and 6-year periods. Click 'Continue' as required until the data is successfully downloaded to your local machine (*data/source_data/ipeds/gradrateraw.csv*). **Data Processing** - *data_cleaning/grad_rates_panel/gradratespanel.R* merges the data together.	data/clean_data	National Center for Education Statistics (2022*b*)	TRUE
IPEDS—Completion	completion.csv	**Data source** (Link): - IPEDs Survey: Completions - IPEDs Title: Awards/degrees conferred by program (6-digit CIP code), award level, race/ethnicity, and gender: (2009-2019). **Data Processing** - *data_cleaning/ipeds_downloading/IPEDS_scraping.R* downloads the data. - *data_cleaning/completion_cleaning/compl_cleaning.R* merges the data together.	data/clean_data	National Center for Education Statistics (2022*a*)	TRUE
PEDS— Tuition revenue and retention rates	welfare.csv	**Data source** (Link): 1) Tuition revenue - IPEDs Survey: Finance - IPEDs Title: all of the finance surveys (2009-2019) 2) Retention rates - IPEDs Survey: Fall Enrollment - IPEDs Title: Race/ethnicity, gender, attendance status, and level of student: Fall (2009-2019) **Data Processing** - *data_cleaning/ipeds_downloading/IPEDS_scraping.R* downloads the data. - *data_cleaning/enrollment_cleaning/welfare_cleaning.R* merges the data together.	data/clean_data	National Center for Education Statistics (2022*a*)	TRUE
IPEDS—Admission and test scores	adm.csv	**Data source** (Link): - IPEDs Survey: Admissions and Test Scores - IPEDs Title: Admission considerations, applications, admissions, enrollees and test scores, fall (2009-2019) **Data Processing** - *data_cleaning/ipeds_downloading/IPEDS_scraping.R* downloads the data.	data/clean_data	National Center for Education Statistics (2022*a*)	TRUE

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Table 1 – continued from previous page

Dataset	Data files	Description and processing	Data Location	Citation	Provided
IPEDS—Fall Enrollment, residency	resid_first_enrol.csv	**Data source** (Link): - IPEDs Survey: Fall Enrollment - IPEDs Title: Residence and migration of first-time freshman: Fall (2009-2019) **Data Processing** - *data_cleaning/ipeds_downloading/IPEDS_scraping.R* downloads the data.	data/source_data/ipeds	National Center for Education Statistics (2022*a*)	TRUE
IPEDS— Finance	finance_all.csv finance_fasb.csv finance_gasp.csv finance_private.csv finance_public.csv	**Data source** (Link): - All Finance surveys except "Response status for all survey components" from 2009 to 2019 **Data Processing** - *data_cleaning/ipeds_downloading/IPEDS_scraping.R* downloads the data.	data/source_data/ipeds	National Center for Education Statistics (2022*a*)	TRUE
IPEDS—Institutional Characteristics	df_inst_char.csv df_inst_char2.csv df_inst_char3.csv	**Data source** (Link): - IPEDs Survey: Institutional Characteristics - IPEDS Titles: Directory information (2009-2019) Educational offerings, organization, services and athletic associations (2009-2019) Student charges for academic year programs (2009-2019) **Data Processing** - *data_cleaning/ipeds_downloading/IPEDS_scraping.R* downloads the data.	data/source_data/ipeds	National Center for Education Statistics (2022*a*)	TRUE
IPEDS—12-Month Enrollment	headcounts.csv	**Data source** (Link): - IPEDs Survey: 12-Month Enrollment - IPEDS Titles: 12-month unduplicated headcount by race/ethnicity, gender and level of student:(2009-2019) **Data Processing** - *data_cleaning/ipeds_downloading/IPEDS_scraping.R* downloads the data.	data/source_data/ipeds	National Center for Education Statistics (2022*a*)	TRUE

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Table 1 – continued from previous page

Dataset	Data files	Description and processing	Data Location	Citation	Provided
IPEDS—Fall Enrollment	quality_measures.csv	**Data source** (Link): - IPEDs Survey: Fall Enrollment - IPEDS Titles: Total entering class, retention rates, and student-to-faculty ratio: (2009-2019) **Data Processing** - *data_cleaning/ipeds_downloading/IPEDS_scraping.R* downloads the data.	data/source_data/ipeds	National Center for Education Statistics (2022*a*)	TRUE
BEA, BLS, and Census Data	bls_bea_data.csv census_agesex.csv census_migration.csv	County level data for population, unemployment rate and per capita income from 2009 to 2019: **BLS data source** (Link) **BEA data source** (Link) **Census data source** (Link) **Data Processing** - *data_cleaning/census_pop/census_data.R* downloads the census data. - *data_cleaning/bls_bea/scraping_functions.R* downloads the BLS and BEA data.	data/source_data/controls	(Bureau of Economic Analysis, 2021, Bureau of Labor Statistics, 2021, U.S. Census Bureau, 2021)	TRUE
Other Data	WeedPrice.csv cpi_data.csv cleaned_price_df.csv google_trend_df.csv trends_raw.csv medical_marij medical.csv	Google trends for marijuana related keywords, marijuana prices, and medical marijuana legalization timeline from 2009 to 2019: **Google Trends** (Link) **Price of Weed data** (Link) **Legalization timeline** (Marijuana Policy Project, 2022, Carnevale Associates, 2022) **CPI data** (Link) **Data Processing** - */data_cleaning/google_trend/google_trend_data.R* processes Google Trends data. - */data_cleaning/marijuana_price/marijuana_price_scraping.R* processes priceofweed.com data and retrieving the old data using archive.org/web.	data/source_data/controls	National Center for Education Statistics (2022*a*)	TRUE

Note: To locate the data on the provided link source, users should visit the National Center for Education Statistics

(NCES) website. Once there, they can navigate to the section containing data from the Integrated Postsecondary Education Data System (IPEDS) surveys. These surveys are identified by two key variables: the IPEDS Survey (Survey column) and the IPEDS Survey Title (Title column). Users can find these identifiers listed in Table 1 under the 'Data Source' column for each IPEDS dataset. Note that *data_cleaning/ipeds_downloading/IPEDS_scraping.R* downloads the raw IPEDS data from National Center for Education Statistics (2022*a*) and saves the data in *data/source_data/ipeds*. In *data_cleaning/ipeds_downloading/IPEDS_scraping.R*, there are functions called to construct panel data from each survey. For instance, the *fall_enroll_race()* function is utilized to extract yearly surveys for the first dataset listed in the table, thereby forming fall enrollment panels. All the data is accessed as of June, 2023. The additional raw data, not listed in the provided table, includes: df_completion.csv (refer to the Completion Dataset in the table for data sources), df_enroll_fall_race.csv (refer to the first Fall Enrollment Dataset in the table for data sources), df_adm_act.csv (refer to the Admission and test scores Dataset in the table for data sources), grad-rate-raw.csv (refer to the Residence Dataset in the table for data sources), and resid_first_enrol.csv (refer to the Residence Dataset in the table for data sources). Further documentation is provided in the *programs/data_cleaning/* subfolders.

Computational Requirements

Software Requirements

While the code should run in most computers as it is not computationally expensive, the code was run on a MacBook Pro with the following specifications:

- System: macOS Monterey (version 12)
- Processor: 2.3 GHz 8-Core Intel Core i9 and
- Memory: 16 GB 2667 MHz DDR4

The only software used is R. The code has been run with R version 4.2.2 (2022-10-31). All programs used can be installed by running *install_load_packages.R* which is called in *master.R*. They include the following:

- *tidyverse*
- *fixest*
- *modelsummary*
- *bacondecomp*
- *fwildclusterboot*
- *haven*
- *ggplot2*
- *dplyr*
- *magrittr*
- *did2s*
- *rvest*
- *lubridate*
- *kableExtra*
- *geosphere*
- *rnaturalearthhires*

- *rmapshaper*
- *tigris*
- *sf*
- *ggspatial*
- *rnaturalearth*
- *matrixStats*
- *readxl*
- *scales*
- *urbnmapr*
- *urbnthemes*
- *devtools*
- *maps*
- *magick*
- *ggpattern*
- *grid*
- *groupdata2*
- *tikzDevice*
- *gtrendsR*

Several packages (*pacbacondecomp, gtrendsR, rnaturalearthhires, urbnthemes, urbnmapr*) are loaded from GitHub using either the *devtools* or *remotes* package. The script *install_load_packages.R* is responsible for executing the package loading process from both CRAN and GitHub. For GitHub packages, users should expect to answer some prompt questions to load all the required packages.

Memory and Runtime Requirements

Running *master.R* takes no more than one hour to run, but replicators can replicate each table and figure separately by running only the source of the table or figure of interest. With the exception of spillover related figures, which require computation of distance between institutions locations and treated states borders, each figure or table should run in no more than 10 minutes.

Description of Programs and Instructions to Replicators

The repository has four main folders: programs, figures, tables, and data. programs/fig_tab folder contains all the scripts needed to replicate all the tables and figures. in programs/fig_tab folder, master.R (orchestrator script) replicates all the tables an figures. The tables are saved in tables folder; figures are saved in figures folder.

programs/data_cleaning contains codes for downloading and saving raw data from the source. For instance, programs/data_cleaning/ipeds_downloading contains code that downloads IPEDS surveys from the NCES source (see ReadMe.Docx file in the same folder). The raw data is then saved in data/source_data/ipeds and the cleaned data which is merged with control variables is then saved in data/clean_data (see scripts in programs/data_cleaning/enrollment_cleaning folder).

Lines 16 to 39 in the *master.R* file execute all data cleaning processes in the specified order. It is important to note that these processes are commented out, so the user needs to uncomment these lines before executing. Additionally, executing this will update all data files. It is worth mentioning that all data was last accessed as of December 2023. As mentioned in footnote 1, Prior to using *bea.R* package to extract BEA data, the user must obtain an API key (see https://github.com/us-bea/bea.R). Starting from line 46, *master.R* runs all the analysis to generates tables and figures on this paper. Please note that main tables presented in the paper's main findings adhere to a 5 percent significance level. Therefore, any indication of significance at the 10 percent level, which is denoted by a plus sign in some main tables, is omitted. For instance, any plus signs in *tab_2_med_completion_Associate's degree.tex* are presented in Table 2 without the "+" sign.

Any minor discrepancies in the wild bootstrap p-values in the tables are solely attributed to the use of different seeds in older versions. Changing the seed in *data-sources.R* will lead to slightly different wild bootstrap p-values. Parallel computing and random number generation algorithms could also introduce slight variations in the wild bootstrap p-values. However, it is important to note that the results remain qualitatively the same despite these differences.

List of Tables, Figures and Programs

To facilitate the replication and comprehension of the code, each script in the "programs/fig_tab" directory is named after the tables or figures it generates. Figures and tables are saved with numerical initials corresponding to their respective numbers in the paper. Figures are stored in either the "figures/main" or "figures/appendix" directory, while tables are saved in the "tables/main" or "tables/appendix" directory.

For example, executing source(*figure_2.R)* in *master.R* generates *Figure 2*, which is then saved as *fig_2_a_main_did.eps* and *fig_2_b_main_did.eps* in the *figures/main* directory. All source code files included in master.R adhere to this convention for creating tables and figures.

References

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